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RESEARCH ARTICLE

Exploring Critical Factors of Crowd-Based Requirement Engineering Using Fuzzy DEMATEL-ANP Method

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ABSTRACT The elicitation of software requirements through the crowd has become an effective approach for requirements development. This approach could improve the quality of the requirement elicitation process, which is a daunting requirement engineering activity. Crowd-based requirement engineering (CrowdRE) has proven beneficial for several systems, specifically those involving diverse stakeholders. This study intends to identify and prioritise the critical factors related to the CrowdRE process through a hybrid Multi-Criteria Decision Making (MCDM) method called DEMATEL-ANP. Hence, in the first place, an extensive literature review was carried out to determine important factors that greatly affect the CrowdRE approach. Secondly, amid the recognised factors, only those with high frequency in the literature were considered for further analysis. The study employs MCDM methods due to their ability to rank and evaluate conflicting alternatives based on expert opinion. Among MCDM techniques, Fuzzy DEMATEL-ANP was chosen, where DEMATEL finds the importance, interrelations, and cause & effect of factors, and ANP assesses relative weights. The fuzzy approach was applied to handle the vagueness and uncertainty among expert opinions. The research results are processed with the help of a custom-developed tool that automates the methods and operations. A comparative analysis was performed to evaluate the performance of Fuzzy and conventional DEMATEL-ANP. The results revealed that Competence is the most important factor, followed by feedback, whereas Anonymity is the least significant factor; thus, it can be excluded from the final list of factors. Among the interrelated factors, competence and feedback were correlated factors of CrowdRE. Moreover, the factors are categorised into cause-and-effect groups. The cause group contains largeness, anonymity, and diversity factors and is termed an influencing factor. The effect group consists of feedback, motivation, competence, and collaboration, known as influenced factors.

INDEX TERMS CrowdRE, critical factors, requirement engineering, fuzzy DEMATEL-ANP, factors interrelationship.

I. INTRODUCTION

Requirement engineering (RE) is the most daunting task in the software development life cycle (SDLC). The RE process plays a key role in the successive stages of SDLC and

influences the overall quality of the final product [1]. Analysts have used many traditional approaches to develop requirements. These conventional techniques include interviews, workshops, and ethnography, which are used to elicit requirements, and inspection and walkthroughs are utilised to validate stakeholder requirements. Traditional techniques allow for in-depth problem exploration but become too expensive and time-consuming when applied among many

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stakeholders. CrowdRE enables us to collect and analyze data from an enormous and diverse group of stakeholders called crowds. A crowd is a diverse group of stakeholders, greater in size for group effects to happen when they intermingle [2]. Through crowds, RE has become an integral approach for software requirement development. The blend of techniques to analyze data from the crowd employing text and usage mining, motivational techniques to actively generate further data, and crowdsourcing to validate requirements is termed “Crowd-based Requirement Engineering (CrowdRE). CrowdRE has proven beneficial to diverse systems involving many stakeholders from various fields. Crowdsourcing and CrowdRE are two different concepts; in the former, some individual tasks are outsourced to a group of people who can perform RE tasks, including elicitation, analysis, and validation, without monetary benefits. In the latter approach, crowd members are intrinsically motivated to improve the effectiveness of the CrowdRE approach [2]. However, CrowdRE depends on crowdsourcing because of high user involvement [2]. In a crowd, only stakeholders with the same goal and objectives in a software development project are involved [4].

Multicriteria decision-making (MCDM) is a set of techniques used to rank, compare, and select from different alternatives that indicate several, often conflicting, criteria. These techniques have been applied in various areas of software engineering for tradeoff analysis [1], [3], [4], [5]. CrowdRE employs MCDM mainly to handle the complexity and subjectivity inherent in eliciting, analyzing, and validating requirements from heterogeneous stakeholders, including end users, domain experts, and other interested users. The major challenge in RE is to cope with diverse, conflicting, incomplete, and ambiguous requirement information offered by several stakeholders. The application of MCDM to CrowdRE makes the process more systematic and crystal clear, and it can settle heterogeneous stakeholders’ diverse and conflicting requirements. MCDM helps in requirement prioritisation, conflict resolution, managing resource constraints, and enhancing the overall success and effectiveness of the RE process [6], [7], [8], [9]. The common MCDM techniques applied to CrowdRE by previous research include the Analytic Hierarchy Process (AHP), Technique for Order Preference by Similarity to Ideal Solution (TOPSIS), and Vlse Kriterijumska Optimizacija I Kompromisno Resenje (VIKOR).

A brief analysis of different MCDM methods reveals some insights that paved the way to choose an effective method for this research. TOPSIS is considered computationally efficient as it can compute both positive-ideal and negative-ideal solutions, but it ignores criteria weighting subjectivity unless integrated with AHP or other methods. Although AHP uses pairwise comparisons, making it intuitive for decision makers, this method also becomes ineffective, like TOPSIS, when handling uncertainty and imprecise information due to the inclusion of subjectivity in expert opinions.

However, the fuzzy approach of AHP handles uncertainty but is considered more appropriate for real-world problems with vague or linguistic data [10]. Unlike other methods, VIKOR requires proper normalization techniques; different methods may influence results, thus causing significant changes in the final ranking of factors [11].

The Fuzzy DEMATEL-ANP method provides, for CrowdRE, several advantages that classical MCDM methods, like AHP or TOPSIS, cannot fully offer. Linked with fuzzy logic, it provides efficient treatment in the case of uncertainty and imprecision of crowds’ inputs, making the indications of stakeholders flexible and more realistic than when data is not precise or complete. In the Decision-Making Trial and Evaluation laboratory (DEMATEL), mapping interdependencies and causes and effects related to requirements may not be adequately elicited by the other methods, thus yielding more complete insight into how a set of requirements influences others. Moreover, the DEMATEL method doesn’t usually require a large number of respondents to get expert opinions [11], [12], [13]. Used along with the Analytic Network Process (ANP), it allows one to undertake a better evaluation that includes the direct and indirect relations of requirements from which one should make prioritizations. It resolves conflicting stakeholder preferences, identifies competing requirements, and prioritizes, considering resource constraints and complex decision criteria. In this way, Fuzzy DEMATEL-ANP increases the transparency of decision-making and enhances the quality of the chosen requirements. It ensures that a structured, data-driven process substantiates decisions and adds value to CrowdRE. The research questions addressed in this study are as follows:

(i) Which influential factors related to CrowdRE have been identified? (ii) How can a mechanism for analyzing CrowdRE factors be developed? (iii) How do we determine the importance and interrelationships among factors? (iv) How do we assess selected factors’ relative weight using MCDM methods? (v) How do we evaluate the proposed solution?

To perform RE processes, including elicitation and analysis of the crowd effectively, an understanding of all of the factors affecting and being affected by CrowdRE is required. There are three main components of CrowdRE, which include Crowd (people working on requirements), Crowdsourcer (the entity who manages the crowd), Crowd process (the method/technique utilised by the crowdsource to solicit requirements from the crowd) [5], [6], [7]. The major factors employed in this study include diversity, largeness, motivation, feedback, competence, anonymity, and collaboration. Prior research has failed to validate the CrowdRE processes and identify/map the factors affecting those processes [8]. This study proposed a hybrid approach employing the Analytical Network Process ANP and DEMATEL. The interdependence constructed through ANP does not represent real-world situations. DEMATEL considers the direct and indirect influence of factors, whereas ANP only considers

direct influence. The blend of these two techniques produces efficient results. DEMATEL also performs critical evaluation, and its blend with ANP can help find relative weights [9]. The hybrid approach of DEMATEL-ANP reduces the subjectivity of expert opinions and generates cause-and-effect diagrams using graph theory and mathematics. Research literature highlighted a challenge regarding the application of DEMATEL, which requires automation of its operations for an efficient analysis [5], [10]. To address this challenge, this study also develops a tool for the DEMATEL method and the Fuzzy DEMATEL-ANP approach to facilitate future research.

II. LITERATURE REVIEW

There are certain variations among crowdsourcing, customer-specific RE, and market-driven RE. In crowdsourcing, some tasks are assigned to people who are provided with monetary benefits instead of participating in RE processes. Customer-specific RE works with small groups of people; for small-scale enterprises, Market-driven RE is used to produce large-scale systems such as enterprise systems. It contains a large and diverse crowd [11]. User involvement is difficult because a large crowd is required for market-driven RE [12]. Researchers suggested a multi-model solution to overcome multiple aspects of CrowdRE. Before applying for CrowdRE, the selection of an appropriate crowd is important. Before proceeding further, their personality traits, including creative potential and skills, should be considered. Some CrowdRE issues are discussed in the literature, mostly related to influential factors. These factors include largeness, diversity, competence, feedback, malicious users, motivation, and crowd involvement.

The factors shown in Table 1 are frequently discussed in the research literature. These factors have a significant influence on CrowdRE. Thus, such factors are considered in this study for the relative assessment of weights. The list of all influential factors, along with their corresponding research studies, is shown in Table 1.

TABLE 1. Influential CrowdRE factors.

Factors	Offered in Literature
Largeness	[14], [2], [15], [16], [17], [18], [19], [20], [21]
Diversity	[14], [2], [15], [16], [18], [19], [21]
Anonymity	[14], [2], [15], [16], [17], [22]
Competence	[14], [15], [16], [17], [19], [20], [23]
Feedback	[24], [14], [15], [16], [25], [26], [27], [28], [29], [21]
Motivation	[24], [14], [2], [15], [16], [30], [31], [28], [21]
Opt-out opportunity	[14], [15], [16]
Volunteering	[14], [15], [16], [32], [31]
Collaboration	[14], [15], [16], [20], [23]

In a relevant study, a survey was performed based on a Likert scale questionnaire. Influential crowd factors were included in this survey to analyze their trends. Some tradeoffs

were listed to show how factors affect each other [14]. Some key findings include: (i) A large crowd provides correct and complete requirements. However, management of such a large crowd can be a major overhead. (ii) The CrowdRE process can be laborious if the crowd is diverse, but it can produce innovative requirements. (iii) Anonymity provides genuine requirements, but reliability concerns may exist. (iv) A motivated crowd could produce the right set of requirements. (v) Collaboration of crowd members may provide more insight into the gathered requirements and the task being performed.

One thing was common in all the processes presented in the literature: the crowd was the main focus of all the RE processes. Each process utilizes the crowd power or manipulates the crowd in a particular way to achieve a specific set of requirements. The research listed a few considerations, which are as follows [24]: (a) A diverse crowd can be more beneficial than a large one, as it ensures the involvement of various types of users, resulting in a more representative and appropriate sample. (b) A transparent working environment should allow participants to leave the crowd-based requirements engineering activity anytime. This voluntary participation eliminates forced involvement, ensuring that requirements are contributed only by motivated members, thereby improving their quality. (c) There should be an appreciation, recognition, and reward system for crowd members in place of their volunteer contribution. (d) The crowd RE process should be dynamic rather than static as the requirements evolve [24]. According to [33] The authors presented a tool, Open Research Knowledge Graph (ORKG), to bridge the gap between researchers and the crowd in the context of CrowdRE. They explore the dual role of individuals as both researchers and crowd members, emphasising the potential of the ORKG to facilitate collaborative research, enhance data interoperability, and support the dynamic integration of crowd-sourced knowledge into formal research processes. Another study highlighted the critical role of RE in ensuring that graduates acquire industry-relevant skills. The findings emphasise the need to integrate RE principles, stakeholder collaboration, and modern requirement elicitation techniques into academic training to enhance students’ preparedness for professional roles in software engineering [34]. Recent studies underline the significance of using MCDM techniques to identify and prioritize key challenges in the domain of software engineering [10], [34], [35], [36].

The challenges related to requirements gathering through a crowd are discussed in [14]. It is stated that the distribution of complex tasks among crowds seems challenging as there may be involvement of malicious users, issues related to analysis, and classifying the crowd based on human characteristics. These issues usually generate bad quality requirements. A study investigated the influence of human factors on the requirement elicitation process to enhance the effective implementation of CrowdRE. Two major challenges encountered in this research include cognitive simulation

in crowd-sourced tasks and requirement filtration from a group. The findings indicate that the quality of requirements is affected by the personal traits of the members of a crowd. This process can be automated to maximize benefits. Given the diverse backgrounds of stakeholders, the feedback collected should be systematically analyzed to derive meaningful insights into the requirements [29]. Another study stressed automated methods to classify feedback, checking its reliability and trends over time to measure the extent of addressed and unaddressed issues [30]. Moreover, since the taxonomy of user feedback involves certain dimensions, a systematic process must be defined for the extraction of requirements from the feedback having several dimensions [25]. A model is proposed for the acquisition of user feedback based on a hierarchy of four factors influencing user feedback [29]. A study examined the progress and challenges in CrowdRE, emphasising the role of online platforms in gathering stakeholder inputs for requirement elicitation. It highlights the need for quality over quantity in crowd-sourced ideas and suggests iterative design approaches for effective CrowdRE implementation [37]. In a recent study, the Crowd-based Requirements Elicitation with User Stories (CREUS) method showcases the potential of crowd-based approaches in requirements engineering, particularly in capturing diverse and high-quality user requirements. By enabling direct user participation, organizations can gather valuable insights that might be overlooked in traditional elicitation methods [38].

It is important to establish a method to address the challenges of CrowdRE and enhance its capabilities. There is a need to design a method to assess the factors that were initially gathered. Realising that the inter-relationships and intra-relationships are necessary to analyze their mutual dependencies [16]. The improved process may be used by practitioners to perform the RE activities in a crowd-based environment, leading to the development of high-quality and innovative requirements [18], [30], [32], [40], [41] [42]. Ascertaining the inter-relationships of factors is a complex activity that involves many facets. Multi-criteria decision-making (MCDM) techniques are the most appropriate approach to deal with such complexity. Before applying MCDM, it is indispensable to first identify the most frequently reported factors in the relevant literature. MCDM methods are widely used in various fields. They handle conflicting alternatives and evaluate and rank alternatives depending on expert opinions and the decision maker's strategy [1], [3], [4], [5]. Alternatives are selected by ranking them from best to worst. Several MCDM problems hold different contexts referring to various scenarios. In several professions like financial analysis and project planning, there are certain strategies that need to be decided considering many criteria and choosing the best among them [43]. CrowdRE is also an MCDM problem as it involves many factors, for which there is a need to find interrelationships and relative weights.

Decision Making Trail and Evaluation laboratory (DEMATEL) is an MCDM method that was developed and introduced by the Geneva Battelle Memorial Institute using graphs and matrices to find complex interrelationships between elements [6], [7], [8], [9]. The two major contributions of DEMATEL include establishing interdependent relations of factors using their cause-and-effect distribution and analysis of critical factors using their inter-relationship map (IRM). DEMATEL aids in prioritising key elements that significantly impact the system by quantifying the degree of influence among factors [11], [13]. DEMATEL can be combined with other MCDM methods to handle its limitations, which include subjectivity due to expert opinions, which might interpret relationships differently. Moreover, as the number of factors increases, the pairwise comparisons for the influence matrix increase exponentially. That's why other MCDM methods are incorporated with DEMATEL to find optimal solutions. DEMATEL is a widely used technique in various fields of study [10]. Several variants of DEMATEL are used in these studies. Numerous research studies highlighted that DEMATEL is the best MCDM method to find interrelationships between complex criteria such as CrowdRE factors. A study examined methodologies and applications of the DEMATEL technique and categorised studies into classical, fuzzy, grey, ANP-integrated, and other DEMATEL variants, providing a comprehensive overview of its utility for finding the interrelationship of factors effectively [13]. DEMATEL is widely applied in various fields, for instance, to evaluate the factors of knowledge management, supply chain management, emergency management, human resource management, the aviation industry, and the manufacturing industry, where it assists in identifying and analysing complex connections between factors. In most of the studies, evaluations were performed through the case study method [44], [45], [46], [47]. As the frameworks introduced in various research studies are mainly validated through case studies rather than real-world projects, the findings may be theoretical or based on case studies, limiting their applicability to a wider range of scenarios.

Recent research proposed a quantitative framework based on the Fuzzy AHP method to discover how CrowdRE factors, including time, budget, risk, and scalability, affect the requirement prioritisation process [48]. Though the fuzzy AHP technique helps deal with uncertainty, it relies heavily on expert judgment. The subjectivity in judgments may lead to biases, eventually affecting the reliability of results and generalisation. Secondly, the study focused only on a selected set of CrowdRE factors and overlooked the other crucial factors impacting the requirement prioritisation. Fuzzy DEMATEL is specifically relevant to CrowdRE as it comprehensively evaluates the connection between various factors. Combining fuzzy techniques handles the uncertainty and ambiguity in crowdsourced data, providing a strong framework for decision-making in the RE process where precise data may be deficient.

Combining fuzzy DEMATEL with ANP can greatly augment the RE process in a crowded environment by improving requirements identification and prioritisation in uncertain and complex settings. The combined DEMATEL-ANP approach can effectively model feedback loops and interrelationships among criteria, enhancing the robustness of MCDM processes [49]. In DEMATEL, with its strengths in analysing and representing interdependence between factors and its dealing with inherent vagueness in human judgments, when combined with ANP, can enable proper consideration of interdependencies and feedback between criteria, and thus a proper prioritisation of requirements accurately and dynamically. With its integration, a better decision can be made by properly considering the complexity in a crowd-sourced environment, and a more effective, adaptable, and collaborative RE process can be constructed with proper consideration of multi-faceted requirements and objectives.

III. RESEARCH METHODOLOGY

The methodology used in this research consists of 4 phases (see Fig. 1). In Phase 1, an extensive literature review has been performed to identify and analyze all factors related to CrowdRE. A survey-based questionnaire is designed in phase 2 of this research. This questionnaire uses the Likert scale to record responses to find the relevant importance of the criteria. Experts' Responses were recorded in MS Excel and imported into a custom-built tool for further analysis. The influence of one CrowdRE factor on other factors was analyzed. Resultantly, pair-wise comparison matrices were generated, depicting the direct and indirect influences of factors on each other. In Phase 3, the Fuzzy DEMATEL method with ANP was applied to find relative weights and interrelationships of CrowdRE factors. In the final phase, evaluation is performed to validate the results.

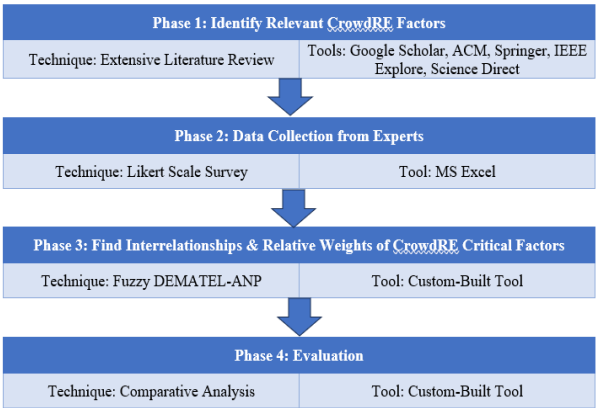


FIGURE 1. Research methodology.

A. PHASE 1: IDENTIFY CROWD FACTORS FOR ANALYSIS

The relevant literature was extracted using automated and manual searches to identify CrowdRE factors. Automated queries are applied to popular research databases: IEEE

Xplore, ACM Digital Library, Springer, and Science Direct. Springer has extracted literature from the computer science and business & management domains. From Google Scholar, literature was retrieved through a manual search. The CrowdRE is diverse and, as of now, less explored, so research studies can be found across different domains of RE beyond the specified keywords. After collecting meta-search terms, customised search strings were formulated for different databases, which are as follows: For Springer, the term requirement should be present with all words; such words may be engineering, elicitation, gathering, and collection, and the title should contain the word “ crowd. For Science Direct, the title, abstract, or keywords should contain words like requirement, whereas the title should contain the word crowd. For IEEE (“paper title”: crowd AND “abstract”: requirement AND (“Index Terms”: crowd* OR (“index terms”: requirement*))). For ACM, the title should match the word ‘crowd’, and the keywords should have the required engineering terms. The research focused on studies that discuss CrowdRE factors and those addressing CrowdRE challenges and presenting their solution. From extensive literature, seven critical factors were identified based on their importance and frequency for further analysis. Some factors were also merged based on their semantics.

B. PHASE 2: DATA COLLECTION

After identifying important CrowdRE factors, their initial impact was analyzed using a questionnaire. The questionnaire consists of 42 questions corresponding to the seven CrowdRE factors. The data collected through the questionnaire was recorded in MS Excel, which was imported into custom-built software for further analysis. In this research, an expert-based evaluation was conducted to evaluate its reliability and effectiveness. The evaluation includes industry professionals and researchers in the CrowdRE domain who gave critical and valuable insights and validation regarding the tool’s applicability and accuracy in decision-making. Moreover, expert evaluation is a widely accepted approach, particularly in the inception stage of research, where direct deployment in a real-world environment may be restricted by time and availability of resources or access to continuing CrowdRE projects. The experts evaluated several aspects of a tool, such as its capability to prioritise factors effectively, the clarity of the interface, and the pertinence of its recommendations. The analysis presents a pairwise comparison matrix. The diagonals of the pair-wise comparison matrix should be zero, indicating that one factor does not have any effect on itself. Each respondent has an individual response matrix depicting his linguistic assessment regarding the effect of factors. This matrix is then processed following the procedure defined for Fuzzy DEMATEL-ANP using a custom-built tool. The survey sample consisted of 33 respondents who belonged to the software industry. The respondents were experts in their field, with 16 years of education in computing and over five years of experience in the software industry. The professionals, including web developers, solution architects, QA engineers,

software project managers, mobile app developers, software engineers, business analysts, and research assistants, all have very good exposure to software requirement engineering processes. Some research assistants were also part of the sample, with 18 years of qualification in computing and working in academia. Their research focused on software requirement engineering and related areas. The participants involved in this study were chosen only from the real software industry collaborating with this project. Global subjects were not available to participate in research at that particular point in time. Table 2 presents the details of the survey sample. Experts' responses were collected and recorded to perform the qualitative research activity further. Most respondents were web developers because they consider various crowd characteristics when designing and developing specific types of websites.

TABLE 2. Details of the survey sample.

Participants	Frequency
Web Developer	31%
Solution Architect	6%
Software QA Engineers	16%
Software Project Manager	16%
Research Assistant	10%
Software Engineer	6%
Mobile Application Developer	9%
Business Analyst	6%

C. PHASE 3: PROPOSED FUZZY DEMATEL-ANP METHOD

DEMATEL only analyses and visualises relationships among factors and does not provide optimal solutions for MCDM problems. It is often combined with other MCDM methods like AHP or ANP. The justification of the method choices employed in this study is depicted in Table 3.

A custom software tool was developed specifically for this research study to find the interrelationships and relative weights of factors. This tool automates the Fuzzy DEMATEL-ANP operations. It is a user-friendly tool; users can set their requirements considering the total number of factors and respondents. It shows the step-by-step execution of operations. The automated execution of the 7×7 matrix, considering the 33 responses gathered through the questionnaire, is shown in Figure 2 below.

1) STEP 1. CONSTRUCT A RESPONSE MATRIX OF EACH EXPERT

The experts gave their opinions through a Likert scale questionnaire specifically collected to satisfy the requirements of this study. The questions were asked regarding the identified factors for CrowdRE, capturing both direct and indirect influences. A fuzzy-based approach was employed to address the inherent vagueness and uncertainty in expert opinions. Linguistic assessment was carried out using five variables (*Extreme Importance*, *Strong Importance*, *Moderate Importance*, *Slight Importance*, and *Equal Importance*). These

TABLE 3. Justification for method choices.

Method	Brief Justification
ANP	- Applied when the decision is based on some interacting elements. Clusters form a network structure connecting elements. Inner and Outer dependence is involved. [52]. - Used to calculate relative weight assessment. - Focus on calculating relationships of factors based on graph theory by generating a relationship map (IRM).
DEMATEL	- Analyse the relations and a relationship map is constructed among particular factors, which is usually based on the category and importance of factors [53]. - Identify both direct and indirect effects of factors. - The dependency of factors on one another is evaluated [12]. - A large number of respondents are not usually required [12]. - Reduces the subjectivity of expert evaluations using a fuzzy approach [53].
Fuzzy DEMATEL-ANP	- DEMATEL is used to find interrelationships among factors. - ANP is used for relative weight assessment of factors depending on their dependency relationships constructed from DEMATEL.

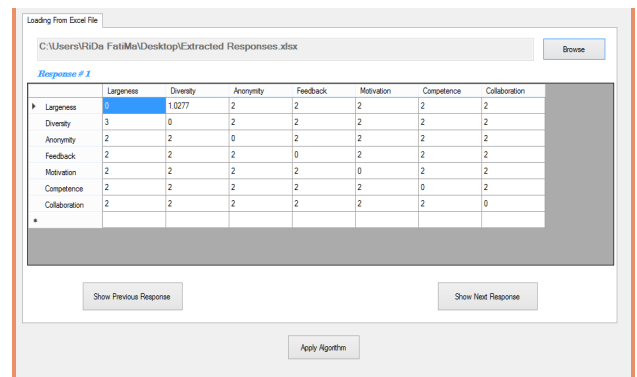


FIGURE 2. Interface of the tool developed for this research work.

linguistic terms were used to gather responses indicating the relative importance of each other. The linguistic variables were then mapped to corresponding Triangular Fuzzy Numbers (TFNs). A linguistic scale, as defined in [13], was used to represent the relative importance of the factors. Table 4 presents the linguistic variables, along with their related TFNs.

TABLE 4. Linguistic scale for relative importance of criteria-fuzzy values.

Linguistic Variable	Triangular Fuzzy Number (TFN) (<i>l, m, u</i>)
Extreme Importance	7/2, 4, 9/2
Strong Importance	5/2, 3, 7/2
Moderate Importance	3/2, 2, 5/2
Slight Importance	2/3, 1, 3/2
Equal Importance	1, 1, 1

The response matrix of each expert is constructed separately, having direct and indirect relationships between factors. Suppose there are total of p respondents and fuzzy matrix $E_p = [E_{ij}^p]_{n \times n}$ is acquired for all experts $E = \{E_1, E_2, E_3, \dots, E_p\}$. The influence of factor F_i on factor F_j is recorded in each cell of the matrix. Seven criteria are represented with a . There will be p number of total fuzzy matrices. These matrices will have linguistic variables at this step, refer to (1). For $i = j$, the diagonal of the matrices is all zero as one factor will not have any effect on itself, refer to (2).

$$E_1 = \begin{bmatrix} 0 & a_{12} & a_{13} & a_{14} & a_{15} & a_{16} & a_{17} \\ a_{21} & 0 & a_{23} & a_{24} & a_{25} & a_{26} & a_{27} \\ a_{31} & a_{32} & 0 & a_{34} & a_{35} & a_{36} & a_{37} \\ a_{41} & a_{42} & a_{43} & 0 & a_{45} & a_{46} & a_{47} \\ a_{51} & a_{52} & a_{53} & a_{54} & 0 & a_{56} & a_{57} \\ a_{61} & a_{62} & a_{63} & a_{64} & a_{65} & 0 & a_{67} \\ a_{71} & a_{72} & a_{73} & a_{74} & a_{75} & a_{76} & 0 \end{bmatrix} \quad (1)$$

$$E_p = \begin{bmatrix} 0 & a_{ij}^p & a_{ij}^p & a_{ij}^p & a_{ij}^p & a_{ij}^p & a_{ij}^p \\ a_{ij}^p & 0 & a_{ij}^p & a_{ij}^p & a_{ij}^p & a_{ij}^p & a_{ij}^p \\ a_{ij}^p & a_{ij}^p & 0 & a_{ij}^p & a_{ij}^p & a_{ij}^p & a_{ij}^p \\ a_{ij}^p & a_{ij}^p & a_{ij}^p & 0 & a_{ij}^p & a_{ij}^p & a_{ij}^p \\ a_{ij}^p & a_{ij}^p & a_{ij}^p & a_{ij}^p & 0 & a_{ij}^p & a_{ij}^p \\ a_{ij}^p & a_{ij}^p & a_{ij}^p & a_{ij}^p & a_{ij}^p & 0 & a_{ij}^p \\ a_{ij}^p & a_{ij}^p & a_{ij}^p & a_{ij}^p & a_{ij}^p & a_{ij}^p & 0 \end{bmatrix} \quad (2)$$

2) STEP 2. CONVERT LINGUISTIC ASSESSMENTS TO TFNS

The linguistic assessments have their corresponding triangular fuzzy numbers to deal with the uncertainty of human judgments. $E_{ij}^p = [e_{ij1}^p, e_{ij2}^p, e_{ij3}^p]$ is the triangular fuzzy number representation of linguistic assessments at step 1. Each cell of the matrix has three elements corresponding to each linguistic term, as shown in (3), as shown at the bottom of the next page. The TFNs set is denoted by (l, m, u) .

3) STEP 3. DEFUZZIFICATION OF EACH RESPONSE MATRIX

The process of getting crisp values from the fuzzy ones is called defuzzification. The matrices obtained up to step 2 contain triangular fuzzy numbers (TFNs). These numbers must be converted into crisp values for further operations. The coordinates of TFNs were denoted as (l, m, u) . Several methods are being used for defuzzification, however, the most widely accepted is the graded mean integration (GMIR) [9], [45], [46].

$$GMIR = \frac{l + 4m + u}{6} \quad (4)$$

Equation (4) is applied to each fuzzy number (l, m, u) shown in step 2. In this way, all initial matrices will be converted to crisp values.

4) STEP 4. CONSTRUCT AN INITIAL DIRECT RELATION MATRIX

In step 3, the individual direct relation matrices contain individual responses from experts. All these responses are

aggregated to obtain a group direct relation matrix. To aggregate responses from all p experts, suppose there are k experts where $1 \leq k \leq p$. The average matrix is constructed by adding all corresponding elements of each matrix referring to expert k and then dividing them by the total number of respondents, p , as referred to in (5).

$$a_{ij} = \frac{1}{p} \sum_{k=1}^p e_{ij}^k \quad (5)$$

D denotes the obtained aggregated direct relation matrix.

5) STEP 5. CONSTRUCT A NORMALISED INITIAL DIRECT RELATION MATRIX

The matrix obtained in step 4 is normalised by calculating the sum of individual rows to get one value for each row to determine the maximum sum. Now, D 's elements are divided by the maximum value of the corresponding row. The obtained values should be between 0 and 1. The obtained normalised matrix is denoted by Z . Suppose n is the total number of elements. Rows are represented by i , and columns are represented by j . S represents the maximum sum.

Equations (6) and (7) represent the steps involved in normalization.

$$S = \frac{1}{\max_{1 \leq i \leq n} \sum_{j=1}^n a_{ij}} \quad (6)$$

where $i, j = 1, 2, 3, \dots, n$

$$Z = D \times S$$

or

$$Z = \frac{D}{\max_{1 \leq i \leq n} \sum_{j=1}^n a_{ij}} \quad (7)$$

where $0 \leq a_{ij} \leq 1$

6) STEP 6. CONSTRUCT TOTAL RELATION MATRIX

Equation (8) represents the Identity matrix I , which is used to formalise the total relation matrix T .

$$T = Z(I - Z)^{-1} \quad (8)$$

7) STEP 7. SUM OF ROWS AND COLUMNS OF TOTAL RELATION MATRIX

The sum of the rows and columns of T is calculated in this step for further processing. Equation (9) shows the sum of rows, denoted by R , and the sum of columns, denoted by C , refer to (10).

$$R = \sum_{i=1}^n a_{ij} \quad (9)$$

$$C = \sum_{j=1}^n a_{ij} \quad (10)$$

where r_i represents both the direct and indirect influences exerted by one factor on several other factors, and c_j represents both the direct and indirect influences a factor receives from various other factors.

The sum prominence is $R+C$, and the difference relation is $R-C$. The factors are distributed among cause-and-effect

groups. Relation is used to distribute factors among these groups. If the corresponding element representing the relation is positive, then factor I will be the causing one and categorised in the cause group, termed the influencing factor. If the relation is negative, factor I will be considered in the effect group and termed an influencing factor.

8) STEP 8. SET THRESHOLD VALUE

The threshold value is set to eliminate all negligible effects. In this way, all the inconsistent responses will be eliminated. DEMATEL generates a direct-relation matrix, followed by applying a threshold to filter out weaker connections between factors. Ignoring factors with negligible influence helps prevent distortion in the final ranking. By setting an appropriate threshold, the method ensures that only the key drivers are considered, resulting in a more accurate prioritisation of critical factors. The average of all elements of the total relation matrix (T) will be the threshold value [8], [52], [53], [54], [55]. All the entries in the matrix that have values less than the corresponding threshold value will be automatically excluded by the decision-maker. Equation (11) represents α which denotes this threshold value, where N is the total number of elements in the matrix.

$$\alpha = \frac{\sum_{j=1}^n \sum_{i=1}^n a_{ij}}{N} \quad (11)$$

9) STEP 9. CONSTRUCT INTER-RELATIONSHIP MAP (IRM) OF FACTORS

To demonstrate the structural relationship between factors, an interrelationship map is constructed to depict the distribution of the factors among the cause-and-effect groups. The horizontal axis plots the $R+C$, and the vertical axis plots the $R-C$. The total relation matrix is the basis of IRM. Map the $(R+C, R-C)$ on the graph to get IRM. If value $a_{ij} > \alpha$ Then, an arrow will be drawn from i to j in the inter-relationship map.

10) STEP 10. CONSTRUCT α -CUT TOTAL RELATION MATRIX

The threshold value computed in step 8 is utilised to obtain the α -cut total relation matrix. All the entries $a_{ij} > \alpha$ remains the same while the values less than α are replaced with zero in the total relation matrix. It is now called the α -cut total relation matrix, referring to (12). T_α denotes the α -cut total

relation matrix.

$$T_\alpha = \begin{Bmatrix} 0 & t_{ij}^\alpha & t_{ij}^\alpha & t_{ij}^\alpha & t_{ij}^\alpha & t_{ij}^\alpha & t_{ij}^\alpha \\ t_{ij}^\alpha & 0 & t_{ij}^\alpha & t_{ij}^\alpha & t_{ij}^\alpha & t_{ij}^\alpha & t_{ij}^\alpha \\ t_{ij}^\alpha & t_{ij}^\alpha & 0 & t_{ij}^\alpha & t_{ij}^\alpha & t_{ij}^\alpha & t_{ij}^\alpha \\ t_{ij}^\alpha & t_{ij}^\alpha & t_{ij}^\alpha & 0 & t_{ij}^\alpha & t_{ij}^\alpha & t_{ij}^\alpha \\ t_{ij}^\alpha & t_{ij}^\alpha & t_{ij}^\alpha & t_{ij}^\alpha & 0 & t_{ij}^\alpha & t_{ij}^\alpha \\ t_{ij}^\alpha & t_{ij}^\alpha & t_{ij}^\alpha & t_{ij}^\alpha & t_{ij}^\alpha & 0 & t_{ij}^\alpha \\ t_{ij}^\alpha & t_{ij}^\alpha & t_{ij}^\alpha & t_{ij}^\alpha & t_{ij}^\alpha & t_{ij}^\alpha & 0 \end{Bmatrix} \quad (12)$$

11) STEP 11. CONSTRUCT NORMALIZED α -CUT TOTAL RELATION MATRIX

There is a need to calculate the normalised α -cut total relation matrix. Take the sum of each T_α column and divide each matrix element by the corresponding column sum, refer to (13). Equation 14 represents the normalised α -cut total relation matrix denoted by T_s .

$$d_i = \sum_{j=1}^n a_{ij}^\alpha \quad (13)$$

$$T_s^\alpha = \begin{Bmatrix} 0 & \frac{t_{ij}^\alpha}{d_i} & \frac{t_{ij}^\alpha}{d_i} & \frac{t_{ij}^\alpha}{d_i} & \frac{t_{ij}^\alpha}{d_i} & \frac{t_{ij}^\alpha}{d_i} & \frac{t_{ij}^\alpha}{d_i} \\ \frac{t_{ij}^\alpha}{d_i} & 0 & \frac{t_{ij}^\alpha}{d_i} & \frac{t_{ij}^\alpha}{d_i} & \frac{t_{ij}^\alpha}{d_i} & \frac{t_{ij}^\alpha}{d_i} & \frac{t_{ij}^\alpha}{d_i} \\ \frac{t_{ij}^\alpha}{d_i} & \frac{t_{ij}^\alpha}{d_i} & 0 & \frac{t_{ij}^\alpha}{d_i} & \frac{t_{ij}^\alpha}{d_i} & \frac{t_{ij}^\alpha}{d_i} & \frac{t_{ij}^\alpha}{d_i} \\ \frac{t_{ij}^\alpha}{d_i} & \frac{t_{ij}^\alpha}{d_i} & \frac{t_{ij}^\alpha}{d_i} & 0 & \frac{t_{ij}^\alpha}{d_i} & \frac{t_{ij}^\alpha}{d_i} & \frac{t_{ij}^\alpha}{d_i} \\ \frac{t_{ij}^\alpha}{d_i} & \frac{t_{ij}^\alpha}{d_i} & \frac{t_{ij}^\alpha}{d_i} & \frac{t_{ij}^\alpha}{d_i} & 0 & \frac{t_{ij}^\alpha}{d_i} & \frac{t_{ij}^\alpha}{d_i} \\ \frac{t_{ij}^\alpha}{d_i} & \frac{t_{ij}^\alpha}{d_i} & \frac{t_{ij}^\alpha}{d_i} & \frac{t_{ij}^\alpha}{d_i} & \frac{t_{ij}^\alpha}{d_i} & 0 & \frac{t_{ij}^\alpha}{d_i} \\ \frac{t_{ij}^\alpha}{d_i} & \frac{t_{ij}^\alpha}{d_i} & \frac{t_{ij}^\alpha}{d_i} & \frac{t_{ij}^\alpha}{d_i} & \frac{t_{ij}^\alpha}{d_i} & \frac{t_{ij}^\alpha}{d_i} & 0 \end{Bmatrix} \quad (14)$$

12) STEP 12. CALCULATE WEIGHTED SUPER MATRIX

ANP uses equal weights to calculate a weighted super matrix. Divide each column of the unweighted super matrix by the total number of criteria, resulting in equal weights as each column sum will be equal to one. This is not a practical approach, so DEMATEL is combined with ANP to resolve this issue [7], [13], [56], [57]. The matrix in step 6 is the unweighted super matrix. Multiply the T_s with the unweighted super matrix to get the weighted super matrix. In this way, the limitation of equal weights in ANP is resolved. Equations (15) and (16), as shown at the bottom of the next page, present the basic calculations.

$$W_s = T \times T_s^\alpha \quad (15)$$

W_s represents the matrix below.

$$E_{ij}^p = \begin{Bmatrix} 0 & (l_{ij}, m_{ij}, u_{ij})^p & (l_{ij}, m_{ij}, u_{ij})^p & \dots & \dots & \dots & \dots \\ \dots & 0 & \dots & \dots & \dots & \dots & \dots \\ \dots & \dots & 0 & \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & 0 & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots & 0 & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots & 0 & \dots \\ \dots & \dots & \dots & \dots & \dots & (l_{ij}, m_{ij}, u_{ij})^p & (l_{ij}, m_{ij}, u_{ij})^p & 0 \end{Bmatrix} \quad (3)$$

α : NUMERICAL EXAMPLE DEMONSTRATING THE PROPOSED SCHEME

This example demonstrates the step-by-step numerical process of the proposed method, Fuzzy DEMATEL-ANP. Three responses were collected for this execution, each involving three factors.

The responses obtained from the questionnaire are represented using TFNs, with all diagonal values set to zero.

$$\text{Response of Respondent 1} = \begin{Bmatrix} 0 & \frac{7}{2}, 4, \frac{9}{2} & \frac{3}{2}, 2, \frac{5}{2} \\ \frac{2}{9}, \frac{1}{4}, \frac{2}{7} & 0 & \frac{2}{3}, 1, \frac{3}{2} \\ \frac{2}{5}, \frac{1}{2}, \frac{2}{3} & \frac{2}{3}, 1, \frac{3}{2} & 0 \end{Bmatrix}$$

$$\text{Response of Respondent 2} = \begin{Bmatrix} 0 & \frac{5}{2}, 3, \frac{7}{2} & 1, 1, 1 \\ \frac{2}{7}, \frac{1}{3}, \frac{2}{5} & 0 & \frac{7}{2}, 4, \frac{9}{2} \\ 1, 1, 1 & \frac{2}{9}, \frac{1}{4}, \frac{2}{7} & 0 \end{Bmatrix}$$

$$\text{Response of Respondent 3} = \begin{Bmatrix} 0 & \frac{2}{3}, 1, \frac{3}{2} & \frac{5}{2}, 3, \frac{7}{2} \\ \frac{2}{3}, 1, \frac{3}{2} & 0 & \frac{3}{2}, 2, \frac{5}{2} \\ \frac{2}{7}, \frac{1}{3}, \frac{2}{5} & \frac{2}{5}, \frac{1}{2}, \frac{2}{3} & 0 \end{Bmatrix}$$

Defuzzification of the above response matrices will be performed in the next step using the formula given in Equation 4. Matrices after defuzzification are shown below

$$\text{Response of Respondent 1} = \begin{Bmatrix} 0 & 4 & 2 \\ 0.2513 & 0 & 1.0277 \\ 0.511 & 1.0277 & 0 \end{Bmatrix}$$

$$\text{Response of Respondent 2} = \begin{Bmatrix} 0 & 3 & 1 \\ 0.336 & 0 & 4 \\ 1 & 0.2513 & 0 \end{Bmatrix}$$

$$\text{Response of Respondent 3} = \begin{Bmatrix} 0 & 1.0277 & 3 \\ 1.0277 & 0 & 2 \\ 0.336 & 0.511 & 0 \end{Bmatrix}$$

All of the above response matrices obtained after defuzzification are aggregated to form a group response matrix. The sum of all the corresponding elements of the three matrices is divided by the total number of respondents. The initial direct relation matrix D obtained is

$$D = \begin{Bmatrix} 0 & 2.6759 & 2 \\ 0.2583 & 0 & 2.3425 \\ 0.6156 & 0.5966 & 0 \end{Bmatrix}$$

Normalising the matrix D by calculating the sum of each row. Identify the maximum row sum and divide each element of the matrix by this maximum value. The resulting normalised

matrix denotes Z .

$$Z = \begin{Bmatrix} 0 & 0.572 & 0.427 \\ 0.115 & 0 & 0.500 \\ 0.131 & 0.127 & 0 \end{Bmatrix}$$

The total relation matrix T is calculated following the procedure outlined in step 6 of the proposed method. The identity matrix I is as:

$$I = \begin{Bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{Bmatrix}$$

$$I-Z = \begin{Bmatrix} 1 & -0.572 & -0.427 \\ -0.115 & 1 & -0.500 \\ -0.131 & -0.127 & 1 \end{Bmatrix}$$

$$(I-Z)^{-1} = \begin{Bmatrix} 1.2145 & 0.8121 & 0.9246 \\ 0.2340 & 1.2243 & 0.7121 \\ 0.1888 & 0.2618 & 1.2115 \end{Bmatrix}$$

$$T = Z(I-Z)^{-1} = \begin{Bmatrix} 0.2144 & 0.8120 & 0.9246 \\ 0.2340 & 0.2242 & 0.7120 \\ 0.1888 & 0.2618 & 0.2115 \end{Bmatrix}$$

Calculate the average of the total relation matrix T , refer to 11, to obtain the threshold value α , which is 0.4203. Elements greater than the threshold will remain unchanged, while those below the threshold will be set to zero. The resulting α -cut total relation matrix is:

$$T_{\alpha} = \begin{Bmatrix} 0 & 0.8120 & 0.9246 \\ 0 & 0 & 0.7120 \\ 0 & 0 & 0 \end{Bmatrix}$$

The values of prominence and relation are determined as follows (refer to equations (9) and (10)).

Prominence (R+C)	Cause/Effect (R-C)
1.7366	1.7366
1.524	-0.1
1.6366	-1.6366

R-C is positive for factor 1, so it will be in the cause group, while factors 2 and 3 have negative values for R-C, thus contributing to the effect group.

To obtain the normalised α -cut total relation matrix, first compute the sum of each column of the α -cut total relation matrix (refer to equation (13)). Then normalise each element

$$W_s = \begin{Bmatrix} 0 & t_{ij}^s \times W_{ij} & t_{ij}^s \times W_{ij} & t_{ij}^s \times W_{ij} & t_{ij}^s \times W_{ij} & t_{ij}^s \times W_{ij} & t_{ij}^s \times W_{ij} \\ t_{ij}^s \times W_{ij} & 0 & t_{ij}^s \times W_{ij} & t_{ij}^s \times W_{ij} & t_{ij}^s \times W_{ij} & t_{ij}^s \times W_{ij} & t_{ij}^s \times W_{ij} \\ t_{ij}^s \times W_{ij} & t_{ij}^s \times W_{ij} & 0 & t_{ij}^s \times W_{ij} & t_{ij}^s \times W_{ij} & t_{ij}^s \times W_{ij} & t_{ij}^s \times W_{ij} \\ t_{ij}^s \times W_{ij} & t_{ij}^s \times W_{ij} & t_{ij}^s \times W_{ij} & 0 & t_{ij}^s \times W_{ij} & t_{ij}^s \times W_{ij} & t_{ij}^s \times W_{ij} \\ t_{ij}^s \times W_{ij} & t_{ij}^s \times W_{ij} & t_{ij}^s \times W_{ij} & t_{ij}^s \times W_{ij} & 0 & t_{ij}^s \times W_{ij} & t_{ij}^s \times W_{ij} \\ t_{ij}^s \times W_{ij} & t_{ij}^s \times W_{ij} & t_{ij}^s \times W_{ij} & t_{ij}^s \times W_{ij} & t_{ij}^s \times W_{ij} & 0 & t_{ij}^s \times W_{ij} \\ t_{ij}^s \times W_{ij} & t_{ij}^s \times W_{ij} & t_{ij}^s \times W_{ij} & t_{ij}^s \times W_{ij} & t_{ij}^s \times W_{ij} & t_{ij}^s \times W_{ij} & 0 \end{Bmatrix} \quad (16)$$

by dividing it by the corresponding column sum.

$$T_{\alpha-\text{nor}} = \begin{Bmatrix} 0.3364 & 0.6255 & 0.5002 \\ 0.3672 & 0.1727 & 0.3852 \\ 0.2962 & 0.2016 & 0.1144 \end{Bmatrix}$$

The weighted super matrix is obtained by multiplying the normalised α -cut total relation matrix with the unweighted super matrix, which is the total relation matrix (refer to equation (15)). The result is as follows.

$$T_s = \begin{Bmatrix} 0.6441 & 0.4607 & 0.5257 \\ 0.3719 & 0.3286 & 0.2848 \\ 0.2222 & 0.2059 & 0.2194 \end{Bmatrix}$$

D. PHASE 4: EVALUATION OF MODEL

In this phase, the validation of the proposed model will be performed. Industry experts determined the relationship between factors in an earlier phase, enhancing the quality of the resulting relationships. However, subjectivity persists owing to data gathering based on human judgment that can be tackled using triangular fuzzy numbers. One type of evaluation is internal to the proposed model, which considers a threshold value to limit the factors. By applying a threshold, only the most significant relationships are considered. This threshold is calculated by taking the average of all elements in the total relation matrix (refer to equation (11)). For further insight, a comparative analysis will be performed to compare the findings of the fuzzy approach, which uses TFNs, and the conventional DEMATEL-ANP, which uses crisp values directly gathered from experts. The same methodology will be used for both approaches, and then the results will be compared to obtain the final weight assessment and identify relationships among factors. We gathered an initial list of factors based on their frequency and importance in the literature. Furthermore, the initial ranking of factors will be compared with the findings to provide a clearer understanding of the results.

IV. RESULTS AND DISCUSSION

A. IDENTIFICATION OF CROWDRE RESEARCH FACTORS THROUGH LITERATURE REVIEW

The initial set of factors identified through the literature review is depicted in Figure 3. It is widely known that not all factors do not necessarily hold high importance [47]. After removing low-importance factors, the final set of CrowdRE factors and their corresponding frequencies are shown in Figure 4. The order of CrowdRE factors from higher to lower significance, along with their definitions, is presented in Table 5.

B. APPLICATION OF FUZZY DEMATEL-ANP METHOD

The response matrix based on the responses gathered through a questionnaire was developed separately for each expert, presenting both direct and indirect relationships between factors. Crisp values are converted to specified fuzzy numbers. The graded mean integration is applied for the defuzzification of crisp values using equation (4) mentioned in the

TABLE 5. Selected factors and their definitions for clarity.

Sr.	Factor	Definition
1	Diversity	Different diversity aspects: age, gender, spatial and expertise, etc.
2	Largeness	Refers to the total number of crowd members, i.e. crowd size.
3	Motivation	There can be intrinsic motivation, including appreciation, or extrinsic motivation, including salary, etc.
4	Feedback	Feedback may or may not be provided to the crowd about their contribution and the following results of the performed activity.
5	Competence	This refers to the skills and creativity of crowd members.
6	Anonymity	Crowd members are unknown to each other.
7	Collaboration	Refers to the communication among crowd members.

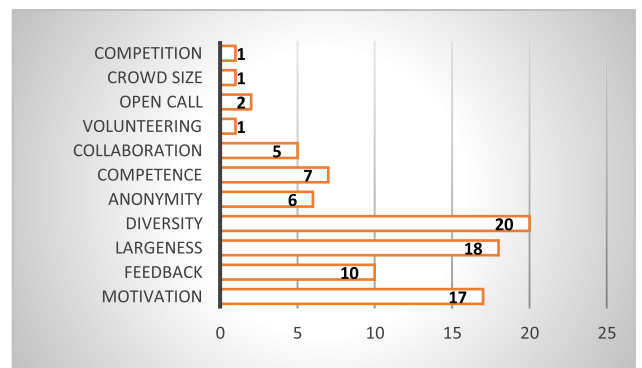


FIGURE 3. The initial set of CrowdRE factors.

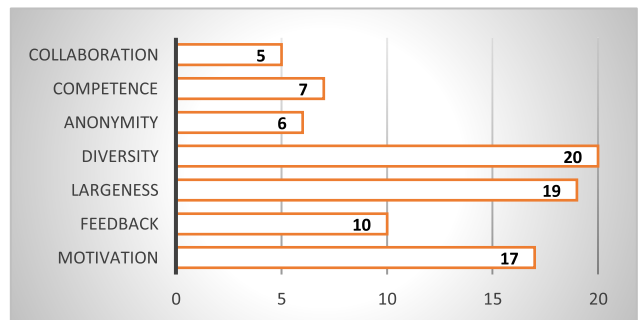


FIGURE 4. Selected CrowdRE factors.

research methodology section. The final values are presented in Table 6.

TABLE 6. TFNs after defuzzification.

Linguistic Variable	Corresponding TFNs after Defuzzification
Extreme Importance	4
Strong Importance	3
Moderate Importance	2
Slight Importance	1.0277
Equal Importance	1

The individual direct relation matrices obtained contain expert responses. These responses are aggregated to obtain a group direct relation matrix (as shown in matrix 1), which is calculated using equation (17), as shown at the bottom of the page, mentioned in the research methodology section. The initial direct relation matrix is normalised by calculating the summation of each row and identifying the maximum sum. Now, elements of the initial matrix are divided by this maximum value. The values obtained from the normalised initial direct relation matrix should be between 0 and 1, as described in the methodology section of the proposed approach and represented here in matrix 2. Table 7 presents Prominence (R+C) and Cause/Effect (R-C). Prominence refers to the degree of importance of factors, while R-C will address the cause-and-effect distribution of factors. This distribution is depicted in Fig. 5. This illustrates the division of factors in the respective groups of cause and effect. The positive value of R-C above the axis represents the cause factors. These factors include largeness, anonymity, and diversity. These are considered influencing factors and need more focus while using CrowdRE. The factors in the effect group are below the axis, which include feedback, motivation, competence, and collaboration. Prominence represents the importance of factors by highlighting their degree. The high prominence value (R+C) factors will be considered most important. It can be seen in Fig. 6 that competence has a high prominence; thus,

it is the most important factor to consider for Crowd-Based Requirements Engineering.

The identity matrix I is used to construct the total relation matrix (as shown in matrix 3), calculated using equation (19), as shown at the bottom of the page, mentioned in the research methodology section.

Anonymity is the least important, having a low value for prominence among all other factors. The factors in the decreasing order of their prominence values are Competence, Feedback, Motivation, Collaboration, Diversity, Largeness, and Anonymity. It can be concluded from these results that competence needs more focus when executing projects involving crowds.

The threshold value is set to eliminate all negligible effects. In this way, all the inconsistent responses will be eliminated. Take the average of all elements in the total relation matrix to get the threshold value.

All the matrix entries having a value less than this calculated threshold value must be excluded. 2.814807 is the threshold value used to obtain the α -cut total relation matrix. All the entries greater than α remain as it is, while the values less than α are replaced with zero in the T. It is now called the α -cut total relation matrix, as shown in matrix 4, calculated using equation (20), as shown at the bottom of the next page, mentioned in the research methodology section. The α -cut total relation matrix will be used to find interrelationships

$$\begin{bmatrix} \text{Factors} & \text{Largeness} & \text{Diversity} & \text{Anonymity} & \text{Feedback} & \text{Motivation} & \text{Competence} & \text{Collaboration} \\ \text{Largeness} & 0 & 2.819861 & 2.515991 & 2.789558 & 2.580794 & 2.819861 & 2.577436 \\ \text{Diversity} & 2.366155 & 0 & 2.457903 & 2.4276 & 2.666667 & 2.880466 & 2.489885 \\ \text{Anonymity} & 2.12373 & 2.488206 & 0 & 2.669185 & 2.547973 & 2.455385 & 2.457064 \\ \text{Feedback} & 2.762612 & 2.487367 & 2.305548 & 0 & 2.758415 & 2.578276 & 2.607739 \\ \text{Motivation} & 2.668346 & 2.367833 & 2.368673 & 2.608579 & 0 & 2.609418 & 2.486527 \\ \text{Competence} & 2.398136 & 2.519348 & 2.518509 & 2.81986 & 2.6069 & 0 & 2.548812 \\ \text{Collaboration} & 2.579115 & 2.397297 & 2.33753 & 2.578276 & 2.547133 & 2.668345 & 0 \end{bmatrix} \quad (17)$$

$$\begin{bmatrix} \text{Factors} & \text{Largeness} & \text{Diversity} & \text{Anonymity} & \text{Feedback} & \text{Motivation} & \text{Competence} & \text{Collaboration} \\ \text{Largeness} & 0 & 0.1751086 & 0.1562388 & 0.1732268 & 0.1602629 & 0.1751086 & 0.1600544 \\ \text{Diversity} & 0.1469342 & 0 & 0.1526316 & 0.1507498 & 0.1655955 & 0.1788721 & 0.1546176 \\ \text{Anonymity} & 0.13188 & 0.1545134 & 0 & 0.1657518 & 0.1582248 & 0.1524752 & 0.1525795 \\ \text{Feedback} & 0.1715535 & 0.1544612 & 0.1431706 & 0 & 0.1712929 & 0.1601065 & 0.1619362 \\ \text{Motivation} & 0.1656997 & 0.1470384 & 0.1470905 & 0.1619883 & 0 & 0.1620404 & 0.1544091 \\ \text{Competence} & 0.1489202 & 0.1564472 & 0.1563952 & 0.1751085 & 0.1618841 & 0 & 0.1582769 \\ \text{Collaboration} & 0.1601587 & 0.1488681 & 0.1451567 & 0.1601066 & 0.1581727 & 0.1656997 & 0 \end{bmatrix} \quad (18)$$

$$\begin{bmatrix} \text{Factors} & \text{Largeness} & \text{Diversity} & \text{Anonymity} & \text{Feedback} & \text{Motivation} & \text{Competence} & \text{Collaboration} \\ \text{Largeness} & 2.546515 & 2.749199 & 2.645832 & 2.877375 & 2.843894 & 2.905664 & 2.751936 \\ \text{Diversity} & 2.692346 & 2.619365 & 2.662014 & 2.883622 & 2.873796 & 2.93175 & 2.770897 \\ \text{Anonymity} & 2.596828 & 2.666979 & 2.446535 & 2.807349 & 2.777582 & 2.817416 & 2.68275 \\ \text{Feedback} & 2.83472 & 2.876659 & 2.771635 & 2.88277 & 3.005333 & 3.043721 & 2.899978 \\ \text{Motivation} & 2.804794 & 2.841312 & 2.749528 & 2.992255 & 2.829872 & 3.018363 & 2.866477 \\ \text{Competence} & 2.831514 & 2.891857 & 2.800637 & 3.050848 & 3.01475 & 2.925706 & 2.912997 \\ \text{Collaboration} & 2.720539 & 2.76309 & 2.670176 & 2.906581 & 2.882288 & 2.935967 & 2.651503 \end{bmatrix} \quad (19)$$

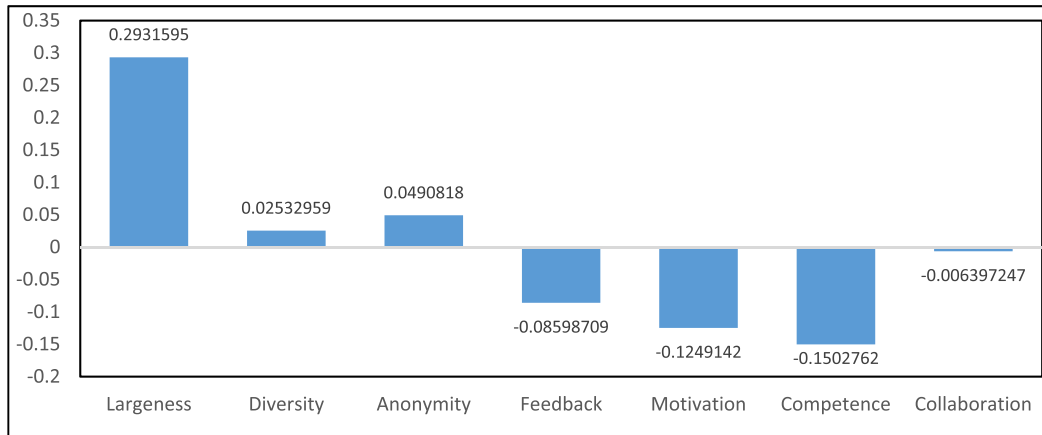


FIGURE 5. Distribution of cause/effect factors using fuzzy values.

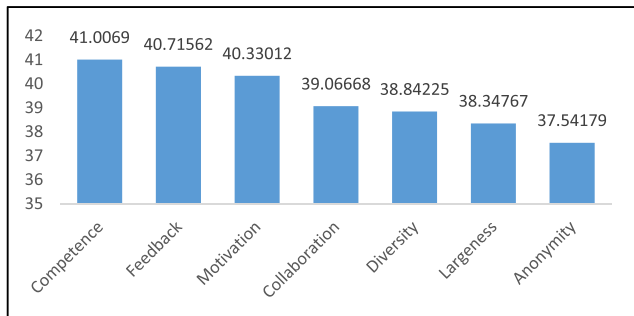


FIGURE 6. Prominence of factors using fuzzy values.

among factors, which clearly shows that anonymity does not affect the performance of Crowd-Based Requirement Engineering. This can also be justified by Fig. 7, which shows the importance of factors where anonymity can be seen as the least important among all other factors. Competence and feedback have the maximum relationships with other factors, which can also be verified from their values of highest prominence.

According to the results, the interrelationship of factors is illustrated using Fig. 7, which indicates that: (i) Anonymity is unrelated to any other CrowdRE factors. (ii) Competence and feedback are the most correlated factors. These factors are interrelated with all other factors. (iii) A single-headed arrow indicates a one-way relationship. There is only one

TABLE 7. Prominence and cause/effect values using fuzzy approach.

Factors	Row Sum - R	Column Sum - C	Prominence (R+C)	Cause/Effect (R-C)
Largeness	19.32042	19.02726	38.34767	0.2931595
Diversity	19.43379	19.40846	38.84225	0.02532959
Anonymity	18.79544	18.74636	37.54179	0.0490818
Feedback	20.31482	20.4008	40.71562	-0.08598709
Motivation	20.1026	20.22751	40.33012	-0.1249142
Competence	20.42831	20.57859	41.0069	-0.1502762
Collaboration	19.53014	19.53654	39.06668	-0.006397247

such relationship where largeness affects motivation, but the reverse is untrue. (iv) Largeness is linked to feedback and competence. (v) Diversity is related to competence, motivation, and feedback. (vi) Motivation is correlated to diversity, feedback, competence, and collaboration.

Next, calculate the normalised α -cut total relation matrix to determine the relative weights of the factors. For each entry in the matrix, divide the sum of each column of the α -cut total relation matrix by the corresponding matrix entry, using the column sum as the divisor. The total relation matrix corresponds to the unweighted super matrix shown in matrix 5, calculated using equation (21), as shown at the bottom of page 15, mentioned in the research methodology section. The weighted super matrix is obtained by multiplying the normalised α -cut total relation matrix with the unweighted super matrix. This approach addresses the limitation of equal weights in ANP.

$$\begin{bmatrix}
 \text{Factors} & \text{Largeness} & \text{Diversity} & \text{Anonymity} & \text{Feedback} & \text{Motivation} & \text{Competence} & \text{Collaboration} \\
 \text{Largeness} & 0 & 0 & 0 & 2.877375 & 2.843894 & 2.905664 & 0 \\
 \text{Diversity} & 0 & 0 & 0 & 2.883622 & 2.873796 & 2.93175 & 0 \\
 \text{Anonymity} & 0 & 0 & 0 & 0 & 0 & 2.817416 & 0 \\
 \text{Feedback} & 2.83472 & 2.876659 & 0 & 2.88277 & 3.005333 & 3.043721 & 2.899978 \\
 \text{Motivation} & 0 & 2.841312 & 0 & 2.992255 & 2.829872 & 3.018363 & 2.866477 \\
 \text{Competence} & 2.831514 & 2.891857 & 0 & 3.050848 & 3.01475 & 2.925706 & 2.912997 \\
 \text{Collaboration} & 0 & 0 & 0 & 2.906581 & 2.882288 & 2.935967 & 0
 \end{bmatrix} \quad (20)$$

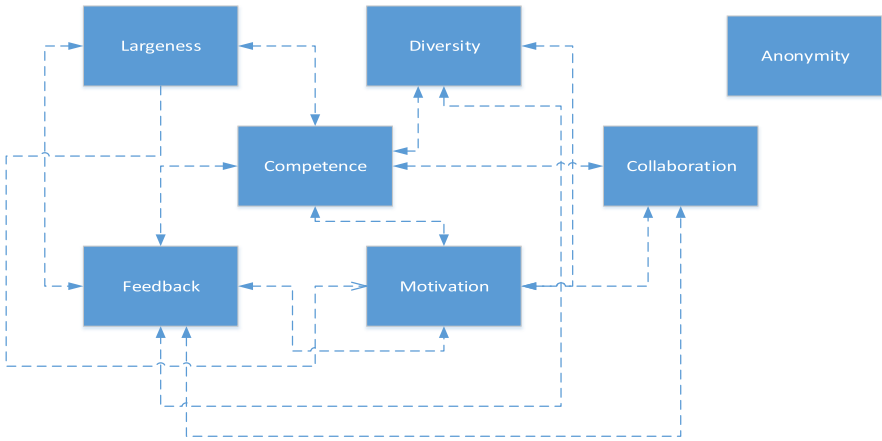


FIGURE 7. Interrelationships of factors using fuzzy values.

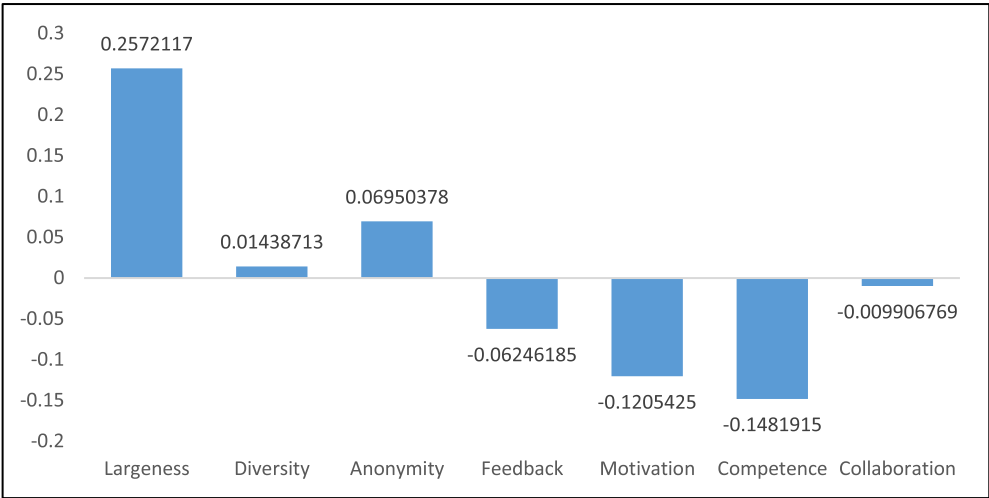


FIGURE 8. Distribution of cause/effect factors using crisp values.

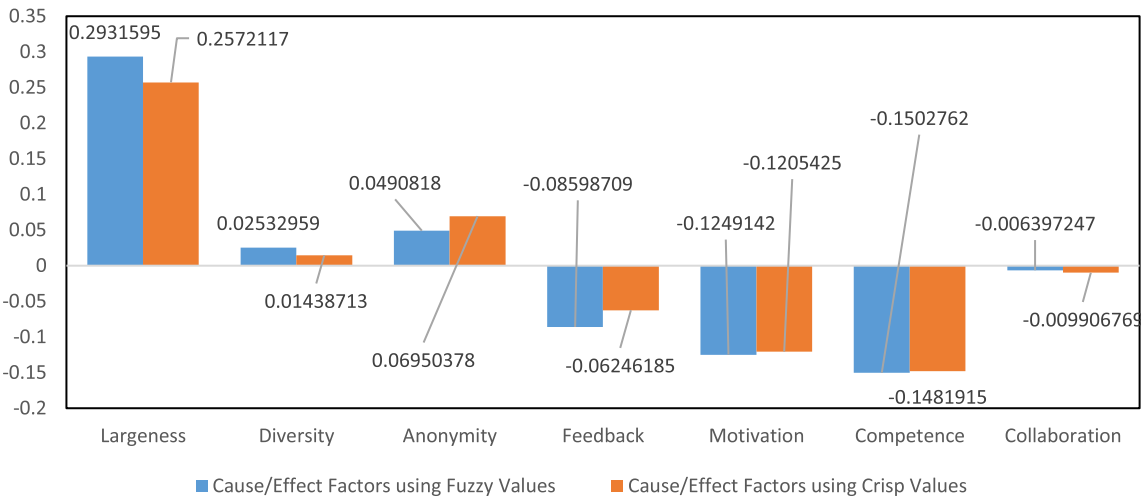


FIGURE 9. Comparison of cause/effect factors using fuzzy and crisp values.

This final weighted super matrix provides the relative weights for the factors, illustrating their interdependencies

as shown in matrix 6, calculated using equation (22), as shown at the bottom of page 15, mentioned in the research

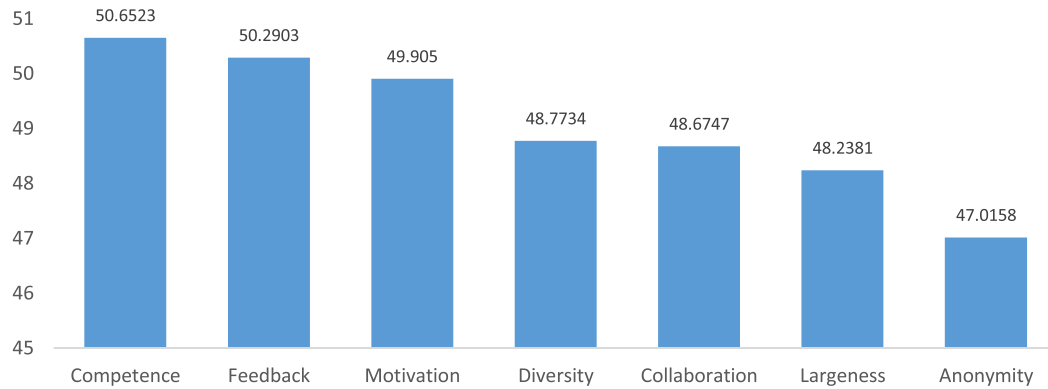


FIGURE 10. The prominence of factors using crisp values.

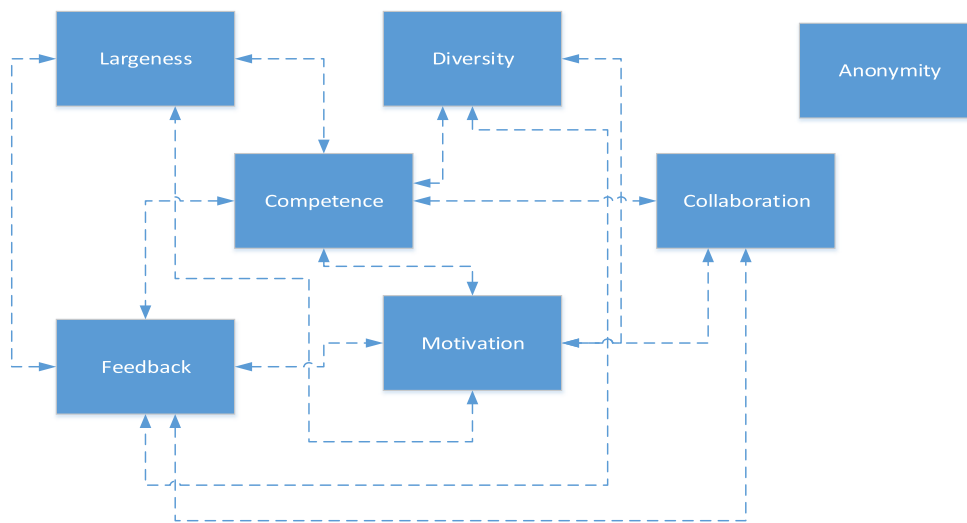


FIGURE 11. Interrelationships of factors using crisp values.

methodology section. (i) Competence and feedback have the highest relative weights, reflecting their prominence in the system. (ii) Anonymity holds the least importance, having the lowest value. (iii) The effect of feedback on competence is the most significant. The second most significant is the effect of competence on feedback.

V. EVALUATION

Experts provided initial relationships of factors through the questionnaire. Further, the proposed model is validated using the calculated threshold value and through comparative analysis to compare results concerning fuzzy and conventional approaches.

A. THRESHOLD VALUE

The DEMATEL technique introduces a threshold value by taking an average of all elements of the total relation matrix, as discussed in the methodology section. In the total relation matrix, elements exceeding this threshold remain unchanged,

while those below it are set to zero. This approach excluded anonymity from the list of factors since all its values in the α -cut total relation matrix were zero, indicating no influence on other CrowdRE factors.

B. COMPARATIVE ANALYSIS

A comparative analysis is performed, where the results of the fuzzy approach using TFNs are compared with the conventional DEMATEL-ANP method, which directly uses crisp values obtained from experts. The crisp values are presented in Table 8. The individual direct relation matrices represent the experts' responses gathered through a Likert-scale questionnaire during the data collection. These individual responses are then aggregated to form a group direct relation matrix.

It is evident that the division using crisp values (see Fig. 8) appears identical when using TFNs (see Fig. 5). There is a difference in values, but the division of cause-and-effect factors remains the same. Prominence reflects

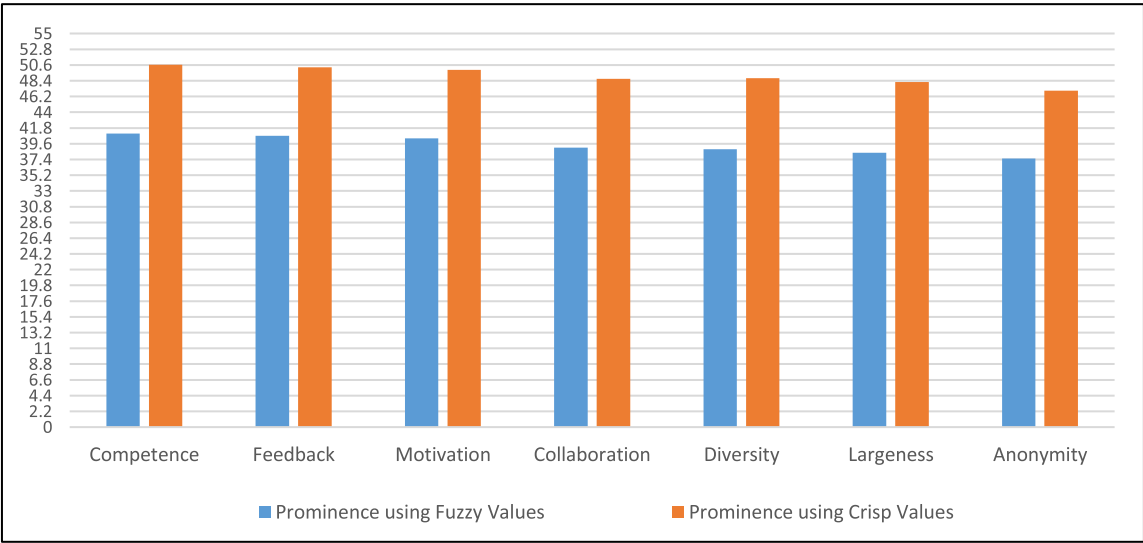


FIGURE 12. Comparison of prominence for fuzzy and crisp values.

TABLE 8. Crisp values.

Linguistic Variable	Crisp Value
Extreme Importance	5
Strong Importance	4
Moderate Importance	3
Slight Importance	2
Equal Importance	1

TABLE 9. Prominence and cause/effect values using crisp values.

Factors	Prominence (R+C)	Cause/Effect (R-C)
Largeness	48.23805	0.2572117
Diversity	48.77342	0.01438713
Anonymity	47.01584	0.06950378
Feedback	50.29027	-0.06246185
Motivation	49.90499	-0.1205425
Competence	50.65234	-0.1481915
Collaboration	48.67471	-0.009906769

the intensity level of importance of certain factors relative to others. While comparing these results with the fuzzy approach, it becomes evident that the order of diversity and collaboration is altered. When using crisp values, diversity is

considered more important than collaboration, and vice versa when using TFNs (see Fig. 9). It is evident that the values for the fuzzy approach (see Table 7) are generally higher

Factors	Largeness	Diversity	Anonymity	Feedback	Motivation	Competence	Collaboration
Largeness	0	0	0	0.163548	0.162974	0.141198	0
Diversity	0	0	0	0.1639032	0.1646881	0.142466	0
Anonymity	0	0	0	0	0	0.136910	0
Feedback	0.500282	0.334113	0	0.163854	0.172226	0.147907	0.334119
Motivation	0	0.330008	0	0.170077	0.162170	0.146674	0.330260
Competence	0.499717	0.335878	0	0.173408	0.172765	0.142172	0.33562
Collaboration	0	0	0	0.165208	0.165174	0.142671	0

(21)

Factors	Largeness	Diversity	Anonymity	Feedback	Motivation	Competence	Collaboration
Largeness	1.320528	1.341859	1.296844	1.389908	1.378391	1.402816	1.352759
Diversity	1.32993	1.351415	1.30609	1.399926	1.388129	1.412778	1.362393
Anonymity	0.3876627	0.395924	0.383435	0.417691	0.412749	0.400558	0.398818
Feedback	4.448855	4.562174	4.447158	4.813047	4.771674	4.865458	4.588171
Motivation	3.139272	3.151147	3.088409	3.334581	3.312531	3.373421	3.175448
Competence	4.468604	4.581818	4.466267	4.832528	4.79241	4.887321	4.608028
Collaboration	1.335575	1.357144	1.311619	1.405769	1.394046	1.418818	1.36817

(22)

compared to the conventional approach that uses crisp values (see Table 9).

The α -cut total relation matrix illustrates the interrelationships among the factors. Anonymity does not impact the performance of CrowdRE when using crisp values. Competence and feedback show strong relationships with other factors, as evidenced by their values of highest prominence.

The interrelationships between the factors depicted in Fig. 11 reveal that anonymity is not related to any other CrowdRE factor, a finding that is also supported by the fuzzy approach. Other key findings include: (a) Competence and feedback exhibit the strongest correlation, which is interrelated with all other factors. (b) Applying the fuzzy approach, it is revealed that largeness does not affect motivation, but not the other way around. However, a bidirectional relationship between largeness and motivation is observed while using crisp values. (c) Diversity is closely related to competence, motivation, and feedback. (d) Similarly, motivation is closely associated with diversity, feedback, competence, and collaboration.

The weighted super matrix shows the interdependence of these factors. Competence, motivation, and feedback show the highest relative weights, reflecting their prominent influence. Anonymity displays zero value, indicating no impact on other factors. Diversity is weighted higher than collaboration, implying a strong influence.

VI. CONCLUSION

Crowd-based approaches have changed the traditional ways of requirement engineering. To effectively perform RE processes using Crowd, a vivid description and categorisation of all the factors of CrowdRE are required. Though factors related to CrowdRE are discussed in existing literature, the importance and dependency relationships among factors are not known, which is very important to successfully conduct CrowdRE. This research study focused on identifying the degree of importance of pertinent factors, their dependencies, and relative weights. To find the interrelationships between factors along with their relative weights, MCDM techniques such as fuzzy DEMATEL-ANP have been applied.

A custom software tool was developed to automatically perform the Fuzzy DEMATEL-ANP operations. The results revealed that competence is the most important factor, and relative weights depict that the effect of feedback on competence has the highest weight. Anonymity ends up being the least important one in terms of prominence value and relative weight. Table 10 presents the factors in descending order, with the first factor of each column showing the highest degree of importance. The results are also compared using fuzzy values and crisp values.

The analysis of the results is summarised as follows: (i) The literature primarily focuses on diversity and largeness, with a very high emphasis frequency. (ii) The results of using Fuzzy DEMATEL-ANP reveal that competence and feedback are the most important factors. (iii) The crisp values yield the same results as the fuzzy approach, except that diversity

TABLE 10. List of factors in descending order of importance, with 1 being the highest.

Sr.	Degree of Importance in Referred Literature	Degree of Importance using Fuzzy Values	Degree of Importance using Crisp Values
7	Diversity	Competence	Competence
6	Largeness	Feedback	Feedback
5	Motivation	Motivation	Motivation
4	Feedback	Collaboration	Diversity
3	Competence	Diversity	Collaboration
2	Anonymity	Largeness	Largeness
1	Collaboration	Anonymity	Anonymity

is more important than collaboration. (iv) Motivation has the same impact overall. (v) Anonymity remains the least important throughout the analysis.

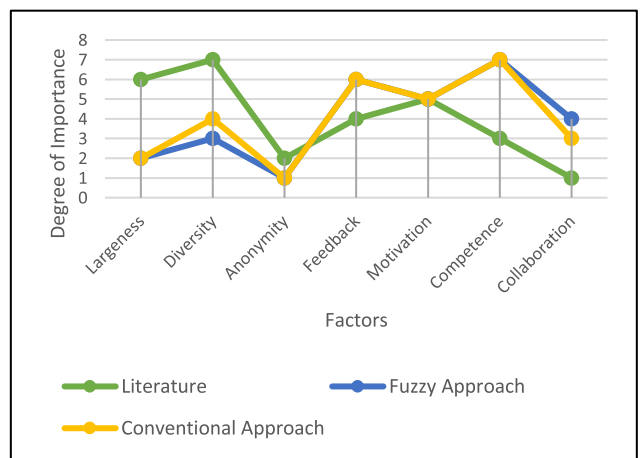


FIGURE 13. Comparative analysis of the degree of importance of factors discussed in the literature, using conventional & fuzzy approaches.

Moreover, the proposed approach has an internal validation process using the threshold value. Values below the specified threshold are discarded, excluding anonymity from the list of factors. The comparative analysis shows that the division of cause-and-effect factors remains consistent across both conventional and fuzzy approaches. However, the order of prominence differs according to diversity and collaboration. Anonymity is consistently ranked as the least important factor. Finally, six factors are critical for CrowdRE, with competence and feedback being the most important. The factors were also categorised into cause-and-effect groups. Largeness, anonymity, and diversity fall under the cause group, as they are considered influential factors that require more attention while using CrowdRE. On the other hand, feedback, motivation, competence, and collaboration belong to the effect group, as the causes influence them.

VII. LIMITATIONS AND FUTURE RESEARCH IMPLICATIONS

RE is the most crippling part of SDLC, which requires utmost effort and attention from the outset of development activity. Ascertaining the quality requirement is a tiresome,

time-consuming, and complicated process. Organizations use CrowdRE to save time and resources while tapping into people with diverse backgrounds and thinking worldwide. However, CrowdRE also produces ambiguous and conflicting requirements, which may lead to the development of an unsuccessful software product. This study attempted to address this problem of CrowdRE by identifying the critical factors, their importance, and their interrelationships through the involvement of relevant experts and MCDM techniques. The research has certain limitations, such as the sample being limited only to the software industry of Pakistan. It only has regional coverage rather than global coverage. The sample mainly consisted of industry professionals rather than equal participation of relevant academic researchers. The selection of participants was influenced by practical constraints, including accessibility to expert professionals and industry collaboration within Pakistan. Future research will enrich the sample by involving participants from academia, industry, and diverse geographical locations to obtain generalizable results.

As far as scalability is concerned, the custom-built tool is currently used to process only seven factors of CrowdRE. Although this tool can process more factors, computational complexity may increase with an increasing number of factors, which will require some efficient data processing techniques. Moreover, an increased number of factors will induce more subjectivity in expert evaluations, which will require consistency checks in order to reduce bias.

In the current study, real-world validation of the tool was not performed owing to certain factors such as access to suitable CrowdRE projects, time and resource constraints. However, the real-world validation of the tool is viewed as an important future step. For future work, the authors aim to collaborate with industry partners to actively engage in CrowdRE and deploy the proposed custom-built tool in real-world projects. Additionally, future research will focus on identifying major dimensions of CrowdRE factors. The proposed approach will be applied to those dimensions and their respective factors. In such a way, the interrelationships among dimensions and factors within each dimension may be calculated for further insight. Furthermore, the study will be augmented with semi-structured interviews with experts from both academia and industry to get richer insights regarding CrowdRE aspects.

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